Open problems:
Unitary evolution as infinitesimal projective processes

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Thermodynamic analogy.** The brief compares the result with the way "reversible quasi-static evolution in thermodynamics can be understood as a limit of equilibration processes". No precise statement linking the two limits is given. (Brief, Sec. 1; Sec. 6: "a reversible quasi-static limit of projective processes".)

2. **Resolvent of an unbounded generator.** The brief states that "direct use of the resolvent of an unbounded generator still gives a completely positive trace-preserving map and the resolvent exponential formula", and that the spectral truncations are needed only for states outside the domain of H. Neither claim is proved, and whether the untruncated $(I - \tau\mathcal{L})^{-1}$ gives perfect prediction on the domain of H is left open. Not formalized. (Brief, Sec. 5, "Unbounded generators"; Log, Sec. 1.1, "Differences", last item.)

From the formalization

Beyond finite dimension the formalization has a part in any dimension, which covers unbounded H in DR-UNIT ⇒ DR-PROJ and derives the evolution structure `IsEvolution` from DR-PROJ up to the item "Durations with irrational ratio" below (`drUnit_iff_drProj_and_map_zero_and_map_add`). It leaves open the item "Resolvent of an unbounded generator" above and the item "General cutoff levels" below.

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **Durations with irrational ratio.** DR-PROJ defines Φ_t only as $\lim_n (\Psi_{t/n})^n$. For t/s irrational a common step τ gives blocks Ψ_τ^k that are not of this form, so "common refinements" and continuity at zero do not prove $\Phi_{t+s} = \Phi_t \circ \Phi_s$; eq. (26) would have to hold uniformly along the trajectory. The remaining interval itself is harmless. A fix: in finite dimension a telescoping estimate gives $\Phi_t = e^{tL}$ directly, which is formalized (`IsProjectiveLimit.exists_eq_exp`); in infinite dimension Chernoff's product formula does, if the closure of L generates a contraction semigroup, which is open. In infinite dimension this is the one clause of `IsEvolution` that is assumed; the others, including composition for durations of rational ratio (eq. (29)), are derived (`IsProjectiveLimit.isEvolution`). $\Phi_0 = I$ is assumed in both dimensions, since the formalization defines Φ_t only for $t > 0$; the brief obtains it from $\Psi_0 = I$.

The gap may be real in infinite dimension. A sketch, not checked in Lean: let $H_1 \neq H_2$ be two self-adjoint extensions of a symmetric operator H_0 with deficiency indices $(1, 1)$, and let Ψ_τ be the cutoff resolvent of H_1 for rational τ and of H_2 for irrational τ. Since t/n is rational exactly when t is, limits, prediction and reconstruction hold for each t, and Φ_t is conjugation by e^{-itH_1} or e^{-itH_2}. If both families have the derivative $[H_0, \rho]/(i\hbar)$ on the density operators built from vectors of $\mathcal{D}(H_0)$, which are dense, then DR-PROJ holds while DR-UNIT would force $H_1 = H_2 + c$. (Brief, Sec. 4, after eq. (29); Log, "Problems found".)

2. [BLOCKS VERIFICATION: BRIEF ISSUE] **What perfect reconstruction means.** DR-PROJ only asks that the probabilities of the reconstructed sequences tend to one. The formalization instead assumes a second projective limit Φ'_t with $\Phi'_t(\Phi_t P) = P$ and $\Phi_t(\Phi'_t P) = P$ on pure states. It is not settled whether the brief's weaker wording already makes Φ_t a bijection of the pure states, or what the right formulation is. (Brief, Sec. 4: "Perfect reconstruction gives the converse correspondence"; Log, "Differences" and "Problems found".)

3. **General cutoff levels.** DR-UNIT $\Rightarrow$ DR-PROJ for unbounded H is formalized for the single choice $M_\tau = \tau^{-1/4}$. The brief's argument for every cutoff satisfying eq. (22) is not formalized. (Brief, eq. (22); Log, Sec. 1.1, "Differences".)

4. **The qubit picture.** $\Psi_\tau = (I + \mathcal{C}_\tau)/2$ with the Cayley transform $\mathcal{C}_\tau = (I + \tau\mathcal{L})(I - \tau\mathcal{L})^{-1}$, its orthogonality for the Hilbert–Schmidt inner product, and the midpoint picture on the Bloch ball are not formalized. (Brief, Sec. 5, "The qubit picture"; Log, "Problems found", last item.)

5. **The finite process is not unitary.** The claims that Ψ_τ is generally irreversible, maps pure ensembles to mixed ones, and that $I - \tau\mathcal{L}$ is not generally positive are not formalized. (Brief, Sec. 5, "The finite process is not unitary"; Log, "Problems found", last item.)